

Geometry

Unit 9

Lesson 7 Homework

Name_____

Date_____

Period_____

1. Oliver lives by the highway. He noticed several tractor trailers drive by every afternoon. Many of the tractor trailers are tankers. The cylinder on a tanker can be 42.6 feet long with a radius of 8.3 feet. What is the maximum volume of the cylinder on a tanker?



2. Oliver counted 16 tanker trucks one Saturday. If every tanker truck was full, what was the total volume of cargo that passed by that day?

3. A baseball has a diameter of 7.5 centimeters. What is the volume of a baseball? Round to the nearest hundredth.



4. The diameter of a softball is 8.89 centimeters. What is the volume of a softball? Round to the nearest hundredth.

5. What is the difference in volume between a baseball and a softball?

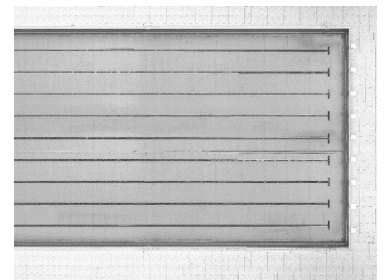
6. Penelope went on a road trip. During the drive, she noticed that some hay bales were rectangular prisms and some were cylindrical. She learned that the first hay bale she saw that was a rectangular prism was 16 inches high by 22 inches wide by 44 inches long. What was the volume of the first hay bale? Round to the nearest hundredth.

7. The second hay bale was cylindrical with a 5 foot diameter and a 4 foot width. What was the volume of the second hay bale, in cubic inches? Round to the nearest hundredth.



8. Which hay bale is larger?

9. The high school swimming pool is 25 meters long, 15 meters wide, and 3 meters deep. What is the volume of water in a full pool?



10. If the pool loses $\frac{1}{10}$ of its water depth during a butterfly swim event, what is the new volume of water?